

Morphological Analysis

Steps in Morphological Analysis for Backpack Design

Problem Description

Clearly define the problem: What type of backpack are you designing? Consider aspects such as purpose (e.g., hiking, school, travel) and target audience.

Identify Dimensions

Break down the design into key dimensions or categories that will influence the backpack's functionality and aesthetics. Common dimensions might include:

Size: Small, medium, large

Material: Nylon, canvas, leather

Style: Casual, formal, sporty

Features: Waterproof, padded straps, multiple compartments

Color Options: Black, blue, red, green

Properties

For each identified dimension, list various properties or options. For example:

Size: 15L, 30L, 50L

Material: Water-resistant nylon, eco-friendly fabric

Style: Backpack with laptop compartment, minimalist design

Features: Hydration reservoir compatibility, reflective strips

Color Options: Solid colors vs. patterns

Combining

Create a Morphological Chart, a matrix that visualizes all possible combinations of the identified dimensions and their properties. This chart allows you to explore various configurations by mixing and matching different options.

Evaluation

Assess the combinations generated in the morphological chart. Evaluate each option for feasibility, functionality, and market appeal. Identify which combinations are practical and which should be eliminated due to incompatibility or lack of desirability.

Implementation

Select the most promising design combinations from the evaluation phase and move forward with prototyping and testing. Gather feedback from potential users to refine the design further.

Conclusion

Using the Morphological Analysis technique allows for a comprehensive exploration of design possibilities for a new backpack. By systematically breaking down the problem into dimensions and properties, designers can generate innovative solutions that meet user needs effectively.

Morphological chart of chair.

Chair components	Physical solutions – S_k				
	1	2	3	4	5
Seat	 S_1 - Square	 S_2 - Half round	 S_3 - Rectangle	 S_4 - Round	 S_5 - Ellipsoidal
Back	 S_6 - Square	 S_7 - Trapezoidal_a	 S_8 - Trapezoidal_b	 S_9 - Round	 S_{10} - Ellipsoidal
Armrest	 S_{11} - "L" shape	 S_{12} - "T" shape	 S_{13} - "J" shape	 S_{14} - Ellipsoidal	 S_{15} - "U" shape
Stand	 S_{16} - Straight_a	 S_{17} - Straight_b	 S_{18} - Round	 S_{19} - Slant_a	 S_{20} - Slant_b